

GenMol UDLM engineering exploration

engineering-v13-ce-temperature-space | complete | 2026-09-07T13:15:46.418543+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

Each temperature-space pair uses the same frozen CE checkpoint and prior. Clean-denoiser temperature is applied before conversion; raw-LOO temperature is applied after conversion. This inference ablation does not add training or backbone evaluations.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED $\geq$ 0.6 and SA $\leq$ 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Predeclared contrasts subtract raw-LOO-temperature from clean-denoiser-temperature within the same seed and decoding branch. Every declared contrast is retained, including negative outcomes. Paired mean and sample SD are withheld unless all declared seed pairs define that metric; unavailable pairs remain visible. Sample SD describes seed spread, not a confidence interval. Pairing seed labels does not imply coupled molecular trajectories.

Prospective sampling design

```
{"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "parameterization_controls": "Both arms use the same CE checkpoint within each prior pair. Control applies temperature after CE-to-LOO conversion; treatment tempers clean logits before conversion and fixes reverse bridge temperature to one. NFE is unchanged.", "predictor_transitions_per_molecule": 128, "randomness": 0.0, "raw_loo_top_p": 1.0, "temperature": 0.5, "temperature_space_comparison": {"primary": {"control_config": "e_ce_raw_t050", "temperature": 0.5, "treatment_config": "e_ce_clean_t050"}, "secondary": {"control_config": "mask_ce_raw_t050", "temperature": 0.5, "treatment_config": "mask_ce_clean_t050"}}, "temperature_space_semantics": {"raw_loo": "loo_conversion_then_temperature_then_reverse_bridge", "x0_denoiser": "clean_denoiser_temperature_then_loo_conversion_then_reverse_bridge_at_one"}, "total_requested_molecule_network_evaluations": 102400}
```

The prospective design was fixed while V12 was running, before reading any MASK-rich molecular outcome. It discloses incidental viewing of one unfinished empirical T1 log. No V12 outcome selected a prior, temperature or excluded arm.

Temperature 0.5 was informed by earlier V5/V6/V9/V10 engineering studies. Both empirical and MASK-rich CE checkpoints are retained; this is a new inference hypothesis, not a bug fix or additional training.

Within each pair only temperature_space changes: same frozen checkpoint, prior, EMA, temperature 0.5, 128 predictor NFE, top-p 1, no Gibbs and other sampling controls. Clean temperature acts before conversion; its reverse bridge temperature is one.

Two seeds of 100 requests are small. Pair by seed, not molecular trajectory. Report every configuration and failure, both strict/repared branches, all four metrics and signed clean-temperature-minus-raw-LOO-temperature contrasts, including negative differences.

Quality counts within-seed unique valid molecules with QED ≥ 0.6 and SA ≤ 4 divided by all requests. Withhold each paired mean/SD unless both declared seed pairs define the metric; SD is descriptive, not a confidence interval.

Disclose final editable MASK and other control counts with the editable-position denominator from sampled_token_control_audit. Decoded strings skip specials and cannot supply these counts; report recovery and component-selection frequencies with denominators.

Training/checkpoint references are inherited unchanged from the accepted V12 protocol; no molecular metrics are used by this preparation. Original V11 failed before training and V8 CT remains a failed controller with a separate checkpoint audit; V11b and V8b are distinct successful follow-ups.

Each CE checkpoint has 1,000 adaptation updates and 128,000 configured exposures from MDLM 50k EMA. These are not necessarily distinct molecules, equal exposures do not imply equal runtime, and final finiteness does not establish every update was finite.

Local MDLM quality 0.858 and paper GenMol V1 quality 0.846 use different training/sample budgets and remain contextual. No superiority, automatic promotion or final-seed evaluation; seeds 0/1/2 remain reserved.

Temperature-space comparison

Both arms use the same frozen clean CE checkpoint within each pair. The control tempers raw LOO logits after denoiser conversion. The treatment tempers clean logits before conversion and uses bridge temperature one. The chosen temperature, prior and all other sampling controls are held fixed within each pair. Neither rule adds a backbone evaluation; this is a new inference hypothesis, not additional training.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"E_CE": {"checkpoint_sha256": "b7d674ed5bddd1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "checkpoint_size_bytes": 1636492999, "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b", "size_bytes": 3161}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}, "udlm_prior_metadata": {"active_token_ids_sha256": "2fd62050919fdf6ee578df881733e47834f662406c219a1f247116e04396a5ae", "active_vocab_size": 1880, "antithetic_sampling": true, "comparison_role": "empirical_prior_treatment", "excluded_token_ids": [], "frequency_active_token_count": 517090, "frequency_artifact_path": "experiments/udlm/token_frequency/train_first_10000.json", "frequency_artifact_schema_version": 1, "frequency_artifact_sha256": "088c78e75611f3cc42c4011e1da6f65a377e673b9cba07a28b126b0fc62f06ed", "frequency_content_token_count": 517090, "frequency_dataset_repo_id": "datamol-io/safe-gpt", "frequency_dataset_revision": "b83175cd7394e7a4027478a35b2f9d1dda3ac62f", "frequency_dataset_selection": "first 10000 streaming rows", "frequency_dataset_split": "train", "frequency_example_count": 10000, "frequency_implementation_git_sha": "56a96b2cd02f9be648a641c51c3d2d8b1ff3033b", "frequency_ordered_text_sha256": "53aee8e5592fc96159788e86519abbbcc9f1ab7c6348a1cb59a939bd57051d8f", "full_vocab_size": 1880, "noise_eps": 0.001, "objective_scope": "model_dependent_ct_integrand_without_parameter_independent_endpoint_kl", "prior_source": "pinned_frequency_artifact_uniform_mixture",

"process_family": "rank_one_continuous_categorical", "sampling_eps": 0.001, "schedule_variant": "schedule_consistent_residual_forward_and_loss", "schema_version": 1, "stationary_probs_sha256": "3b6ce79449e0abc083bc860b3543be554667435ce9f7f1b92c0f960ff27400c3", "tokenizer_json_sha256": "0db5f4dbdc7e8ff759e98483759611a426e187ee7f3f0a91edc8800abe7bf140", "tokenizer_repo_id": "datamol-io/safe-gpt", "tokenizer_revision": "3d5fa0988383e898d5ac5db7cd52bf715bc37061", "uniform_mixture_weight": 0.0002, "variant": "empirical_frequency"}}}, {"MASK_CE": {"checkpoint_sha256": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "checkpoint_size_bytes": 1636493702, "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/launch_manifest.json", "sha256": "457912bcf1975f0adb72cb2b5f02ec07c32915dc2119f184c19d6003edb60f85", "size_bytes": 24826}, "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v11b_mask_prior.json", "sha256": "5ecf638fa8fc7a3707a0497c8a358697610f52c8af7f2d6468458890e26368eb", "size_bytes": 7982}, "request_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/request_manifest.json", "sha256": "a039045f0e3589b3c15ac56fa215cb27bf0bde36b87902c0e1028c456b45f956", "size_bytes": 14003}, "resolved_config_sha256": "5d73af507e81b643cf8f35225ab5d796357024d1665d039fdf40b5f37df15698", "source_revision": "be18a3244a717978e027d6e43ba0ac20b8137e4a", "terminal_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/terminal_manifest.json", "sha256": "891f5d6d609146a02ebb18caa7176bf555e3f67fe2493795c8a3e1fccb264779", "size_bytes": 17667}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}, "udlm_prior_metadata": {"active_token_ids_sha256": "2fd62050919fdf6ee578df881733e47834f662406c219a1f247116e04396a5ae", "active_vocab_size": 1880, "antithetic_sampling": true, "base_stationary_probs_sha256": "3b6ce79449e0abc083bc860b3543be554667435ce9f7f1b92c0f960ff27400c3", "comparison_role": "mask_rich_empirical_prior_treatment", "excluded_token_ids": [], "frequency_active_token_count": 517090, "frequency_artifact_path": "experiments/udlm/token_frequency/train_first_10000.json", "frequency_artifact_schema_version": 1, "frequency_artifact_sha256": "088c78e75611f3cc42c4011e1da6f65a377e673b9cba07a28b126b0fc62f06ed", "frequency_content_token_count": 517090, "frequency_dataset_repo_id": "datamol-io/safe-gpt", "frequency_dataset_revision": "b83175cd7394e7a4027478a35b2f9d1dda3ac62f", "frequency_dataset_selection": "first 10000 streaming rows", "frequency_dataset_split": "train", "frequency_example_count": 10000, "frequency_implementation_git_sha": "56a96b2cd02f9be648a641c51c3d2d8b1ff3033b", "frequency_ordered_text_sha256": "53aee8e5592fc96159788e86519abbbcc9f1ab7c6348a1cb59a939bd57051d8f", "full_vocab_size": 1880, "mask_mixture_weight": 0.9, "mask_token_id": 4, "noise_eps": 0.001, "objective_scope": "model_dependent_ct_integrand_without_parameter_independent_endpoint_kl", "prior_source": "mask_point_mass_mixture_with_pinned_smoothed_frequency", "process_family": "rank_one_continuous_categorical", "sampling_eps": 0.001, "schedule_variant": "schedule_consistent_residual_forward_and_loss", "schema_version": 1, "stationary_probs_sha256": "07d011c07bdac8fdb096f514ded6e2d1b6850268c48bec8dd0e3f3853d013eb", "tokenizer_json_sha256": "0db5f4dbdc7e8ff759e98483759611a426e187ee7f3f0a91edc8800abe7bf140", "tokenizer_repo_id": "datamol-io/safe-gpt", "tokenizer_revision": "3d5fa0988383e898d5ac5db7cd52bf715bc37061", "uniform_mixture_weight": 0.0002, "variant": "mask_rich_empirical"}}}, {"common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_definition": "Empirical-arm configuration after removing parameterization and normalizing callback.dirpath. The MASK arm matches after restoring empirical_frequency and removing mask_mixture_weight.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v11_failure": {"relative_path": "output/udlm/engineering_v11/mask_ce_1000_b128_w2/terminal_manifest.json", "sha256": "6a4aa47e1f7b76ad5efc3ce03cd3e4a55c0db4d95778b0cf60cc55cab7207408", "size_bytes": 14510}, "original_v8_ct_context": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}, {"original_v8_protocol": {"relative_path":

"experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"},
"peak_learning_rate": 0.0003, "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50},
"training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdln_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
e_ce_raw_t050	2100	128	128	0
e_ce_raw_t050	2101	128	128	0
e_ce_clean_t050	2100	128	128	0
e_ce_clean_t050	2101	128	128	0
mask_ce_raw_t050	2100	128	128	0
mask_ce_raw_t050	2101	128	128	0
mask_ce_clean_t050	2100	128	128	0
mask_ce_clean_t050	2101	128	128	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
e_ce_raw_t050	2100	clean CE	x0_denoiser	ema
e_ce_raw_t050	2101	clean CE	x0_denoiser	ema
e_ce_clean_t050	2100	clean CE	x0_denoiser	ema

Config	Seed	Objective	Parameterization	Weights
e_ce_clean_t050	2101	clean CE	x0_denoiser	ema
mask_ce_raw_t050	2100	clean CE	x0_denoiser	ema
mask_ce_raw_t050	2101	clean CE	x0_denoiser	ema
mask_ce_clean_t050	2100	clean CE	x0_denoiser	ema
mask_ce_clean_t050	2101	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_raw_t050	2/2	0.9900 +/- 0.0141	0.9798 +/- 0.0003	0.4650 +/- 0.0212	0.8934 +/- 0.0053
e_ce_clean_t050	2/2	0.9750 +/- 0.0212	1.0000 +/- 0.0000	0.5200 +/- 0.0990	0.8914 +/- 0.0011
mask_ce_raw_t050	2/2	0.9900 +/- 0.0000	0.9949 +/- 0.0071	0.4950 +/- 0.0354	0.8817 +/- 0.0007
mask_ce_clean_t050	2/2	0.9850 +/- 0.0071	0.9949 +/- 0.0072	0.4750 +/- 0.0212	0.8916 +/- 0.0091

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_raw_t050	2/2	0.6650 +/- 0.0495	1.0000 +/- 0.0000	0.3450 +/- 0.0212	0.8777 +/- 0.0022
e_ce_clean_t050	2/2	0.6400 +/- 0.0566	1.0000 +/- 0.0000	0.3450 +/- 0.0636	0.8761 +/- 0.0024
mask_ce_raw_t050	2/2	0.8050 +/- 0.0212	1.0000 +/- 0.0000	0.4200 +/- 0.0283	0.8728 +/- 0.0034
mask_ce_clean_t050	2/2	0.6650 +/- 0.0495	1.0000 +/- 0.0000	0.3250 +/- 0.0071	0.8783 +/- 0.0048

Predeclared paired clean-temperature minus raw-LOO-temperature contrasts

Differences use the same seed and decoding branch, with equal seed weights. Values are on the metric's 0-to-1 scale: 0.01 is one percentage point for quality. All declared pairs must define a metric before its mean and sample SD are shown. SD is seed spread, not a confidence interval. Seed labels do not imply paired molecules.

primary | temperature 0.5: e_ce_clean_t050 minus e_ce_raw_t050

released_comparable

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.980000	+0.960000	-0.020000	completed
validity	2101	+1.000000	+0.990000	-0.010000	completed
uniqueness	2100	+0.979592	+1.000000	+0.020408	completed
uniqueness	2101	+0.980000	+1.000000	+0.020000	completed
quality	2100	+0.450000	+0.450000	+0.000000	completed
quality	2101	+0.480000	+0.590000	+0.110000	completed
diversity	2100	+0.897172	+0.890625	-0.006547	completed
diversity	2101	+0.889607	+0.892243	+0.002636	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.015000	0.007071
uniqueness	2/2	+0.020204	0.000289
quality	2/2	+0.055000	0.077782
diversity	2/2	-0.001956	0.006493

strict

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.630000	+0.600000	-0.030000	completed
validity	2101	+0.700000	+0.680000	-0.020000	completed
uniqueness	2100	+1.000000	+1.000000	+0.000000	completed
uniqueness	2101	+1.000000	+1.000000	+0.000000	completed
quality	2100	+0.330000	+0.300000	-0.030000	completed
quality	2101	+0.360000	+0.390000	+0.030000	completed

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
diversity	2100	+0.879281	+0.874420	-0.004861	completed
diversity	2101	+0.876110	+0.877813	+0.001703	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.025000	0.007071
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.000000	0.042426
diversity	2/2	-0.001579	0.004641

secondary | temperature 0.5: mask_ce_clean_t050 minus mask_ce_raw_t050

released_comparable

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.990000	+0.980000	-0.010000	completed
validity	2101	+0.990000	+0.990000	+0.000000	completed
uniqueness	2100	+1.000000	+0.989796	-0.010204	completed
uniqueness	2101	+0.989899	+1.000000	+0.010101	completed
quality	2100	+0.520000	+0.460000	-0.060000	completed
quality	2101	+0.470000	+0.490000	+0.020000	completed
diversity	2100	+0.881186	+0.898097	+0.016911	completed
diversity	2101	+0.882173	+0.885188	+0.003015	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.005000	0.007071
uniqueness	2/2	-0.000052	0.014358
quality	2/2	-0.020000	0.056569

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
diversity	2/2	+0.009963	0.009826

strict

Metric	Seed	LOO temp	Clean temp	Clean minus LOO	Pair status
validity	2100	+0.820000	+0.630000	-0.190000	completed
validity	2101	+0.790000	+0.700000	-0.090000	completed
uniqueness	2100	+1.000000	+1.000000	+0.000000	completed
uniqueness	2101	+1.000000	+1.000000	+0.000000	completed
quality	2100	+0.440000	+0.320000	-0.120000	completed
quality	2101	+0.400000	+0.330000	-0.070000	completed
diversity	2100	+0.875190	+0.881751	+0.006561	completed
diversity	2101	+0.870420	+0.874920	+0.004500	completed

Metric	Pairs defined	Mean Clean minus LOO	Sample SD
validity	2/2	-0.140000	0.070711
uniqueness	2/2	+0.000000	0.000000
quality	2/2	-0.095000	0.035355
diversity	2/2	+0.005531	0.001458

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_ce_raw_t050	2100	completed	100	100	0.4500	0.3300	17.69
e_ce_raw_t050	2101	completed	100	100	0.4800	0.3600	18.02
e_ce_clean_t050	2100	completed	100	100	0.4500	0.3000	19.79
e_ce_clean_t050	2101	completed	100	100	0.5900	0.3900	20.84

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
mask_ce_raw_t050	2100	completed	100	100	0.5200	0.4400	14.10
mask_ce_raw_t050	2101	completed	100	100	0.4700	0.4000	14.95
mask_ce_clean_t050	2100	completed	100	100	0.4600	0.3200	14.24
mask_ce_clean_t050	2101	completed	100	100	0.4900	0.3300	16.00

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={ 'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected { 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: daf8c8108772e0ebcbdd39f15dab07478fe8945ba6b33f4b68858a8536ad7319

v13-e-ce-raw-t050 / seed 2100: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison":

"strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-raw-t050/seed_2100/raw_samples.csv", "sha256": "f625f31422a820d1b9384b9b3fc38668b42a860f293132ae241de2233509362d", "size_bytes": 32295}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-e-ce-raw-t050 / seed 2101: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-raw-t050/seed_2101/raw_samples.csv", "sha256": "c235b82d57e6f54bdc28b304ec1f6a7c569dcf8b240a0694df52bd7c5727a70e", "size_bytes": 33222}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-e-ce-clean-t050 / seed 2100: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus",

"max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3", "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-clean-t050/seed_2100/raw_samples.csv", "sha256": "598044d70dfcef4bc968f25dbfa7537f407c31806fa64c8003f72950ee3504a5", "size_bytes": 32059}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-e-ce-clean-t050 / seed 2101: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2", "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-e-ce-clean-t050/seed_2101/raw_samples.csv", "sha256": "66b3a97cdbaa2571bb4b6fb3643d42b63f58d922d74947c40609e9e20f69e210", "size_bytes": 33251}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-raw-t050 / seed 2100: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid":

"GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": 3}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-raw-t050/seed_2100/raw_samples.csv", "sha256": "1eb3b11226c43e83db59ecc80c5bacc10051629d88f44c565b172e8d3dfbade9", "size_bytes": 34211, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-raw-t050 / seed 2101: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": 2}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-raw-t050/seed_2101/raw_samples.csv", "sha256": "a872bdcac13280e7f805594abe6eafae6afe1d3b3f4289e0e82035bf60e8517d", "size_bytes": 34038, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-clean-t050 / seed 2100: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python",

"used_memory_mib": 1564}}, "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2", "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-clean-t050/seed_2100/raw_samples.csv", "sha256": "557a800a61fde29b1f4588a35aa923bd7090906aa886e5f8d3c5d2938775aef7", "size_bytes": 32216}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v13-mask-ce-clean-t050 / seed 2101: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "temperature_space": "x0_denoiser"}, "denoiser_source_sha256": "ca3b968b529eb08bb6d4f1ecbbad087ee4a5eae465c316634327746603f2b21b", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLN reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "reverse_bridge_temperature": 1.0, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "temperature_application": "clean_denoiser_before_loo_conversion", "temperature_space": "x0_denoiser"}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3", "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v13/v13-mask-ce-clean-t050/seed_2101/raw_samples.csv", "sha256": "763d698ec814de379f24772cd6c87d3323d921a0cee54f588ad7a711d3beb39c", "size_bytes": 33316}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "0fdc54d91baade2f84b48de9223ceb19696fadb7", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}